



SIMATS ENGINEERING



Saveetha Institute of Medical and Technical Sciences

Chennai 2026

ASSESSMENT TOOL -3

CO2_AT3

Name : N.Sunil Kumar

Register No : 192472330

Course code : CSA0708

Course name: Computer Networks

Faculty name: Dr.S.Saraswathi

Debugging & Optimisation Task

Questions:-

1. A network administrator has configured a workstation with the following settings:

- **IP Address: 192.168.10.65 ;Subnet Mask: 255.255.255.192 (/26) ;Default Gateway: 192.168.10.1**

However, the workstation is unable to communicate with other systems configured in the subnet 192.168.10.0/26.

Task :-Debug the network configuration by:

- **Identifying the subnet to which the configured IP address belongs.**
- **Determining the network address, broadcast address, and valid host range.**
- **Verifying whether the configured IP address and gateway are valid.**
- **Identifying the configuration errors.**
- **Optimizing the configuration by assigning an appropriate IP address and default gateway.**
- **Calculating the total number of usable host addresses after correction.**

ANSWER :-

IPv4 Network Debugging

Given:

- IP Address: 192.168.10.65
- Subnet Mask: 255.255.255.192 (/26)
- Gateway: 192.168.10.1

Step 1: Identify the Subnet

A /26 subnet has blocks of 64.

Subnet	Address Range
192.168.10.0/26	0–63
192.168.10.64/26	64–127
192.168.10.128/26	128–191
192.168.10.192/26	192–255

Step 3: Verify Configuration

Setting: IP 192.168.10.65

Status: Valid

Setting: Gateway 192.168.10.1

Status: Invalid

(The gateway is in 192.168.10.0/26, while the workstation is in 192.168.10.64/26.)

Step 4: Configuration Error

- Wrong default gateway
- Host and gateway are in different subnets

Step 5: Optimized Configuration

- IP Address: 192.168.10.65
- Subnet Mask: 255.255.255.192
- Default Gateway: 192.168.10.126 *(or any valid router IP in this subnet)*

Step 6: Usable Hosts

- Calculation: $2^{(32-26)} - 2 = 64 - 2 = 62$
- Usable Hosts: 62

2. Two devices are configured with the following IPv6 addresses:

- **Device A: 2001:db8::ff00:42:8329/64**
- **Device B: 2001:db8:0:0:1::1/64**

Although both devices are connected to the same network, communication fails.

Task :

Debug the IPv6 configuration by:

- **Expanding both IPv6 addresses into their complete notation.**
- **Determining the network prefix of each device.**
- **Verifying whether both devices belong to the same subnet.**
- **Identifying the addressing or prefix mismatch.**
- **Optimizing the IPv6 configuration by assigning appropriate addresses.**
- **Calculating the number of available host addresses within the corrected subnet.**

ANSWER :-

Given

Device A: 2001:db8::ff00:42:8329/64

Device B: 2001:db8:0:0:1::1/64

Expanded Addresses

- Device A: 2001:0db8:0000:0000:0000:ff00:0042:8329
- Device B: 2001:0db8:0000:0000:0001:0000:0000:0001

Network Prefix

- Device A Prefix: 2001:db8:0:0::/64
- Device B Prefix: 2001:db8:0:0::/64

Same Subnet?

Yes. Both belong to 2001:db8:0:0::/64.

Issue

The addressing is correct; the communication failure is likely due to:

- Incorrect gateway
- Interface disabled
- Duplicate address

- Firewall/Neighbor Discovery issue

Optimized Addresses

- Device A: 2001:db8:0:0::10/64
- Device B: 2001:db8:0:0::20/64

Available Hosts

A /64 subnet provides:

- Calculation: $2^{64} = 18,446,744,073,709,551,616$
- Available Hosts: 2^{64}

3. An organization has allocated the network 192.168.1.0/24 into several departments using the subnet masks /26, /27, and /28. Some departments report insufficient IP addresses, while others have a large number of unused addresses.

Task : Analyze and optimize the subnet allocation by:

- Identifying the network address, broadcast address, and host range for each subnet.
- Calculating the number of usable and unused IP addresses.
- Identifying inefficient subnet allocations.
- Debugging the addressing plan by locating subnet allocation mistakes.
- Recommending an optimized subnetting scheme using VLSM to improve address utilization while supporting current departmental requirements.

ANSWER :-

IPv4 Subnet Optimization (VLSM) Network: 192.168.1.0/24

Existing Allocation

- Subnet /26:
 - Network: 192.168.1.0
 - Host Range: 1–62
 - Broadcast: 63
 - Usable: 62
- Subnet /27:
 - Network: 192.168.1.64
 - Host Range: 65–94
 - Broadcast: 95
 - Usable: 30

- Subnet /28:
 - Network: 192.168.1.96
 - Host Range: 97–110
 - Broadcast: 111
 - Usable: 14

Address Utilization

- Subnet /26:
 - Total: 64
 - Usable: 62
 - Unused: High
- Subnet /27:
 - Total: 32
 - Usable: 30
 - Unused: Moderate
- Subnet /28:
 - Total: 16
 - Usable: 14
 - Unused: Low

Inefficient Allocation

- Small departments using /26 waste addresses.
- Growing departments on /28 may run out of IPs.

Optimized VLSM Scheme

- Admin:
 - Required Hosts: 50
 - New Subnet: 192.168.1.0/26
- Sales:
 - Required Hosts: 25
 - New Subnet: 192.168.1.64/27
- HR:
 - Required Hosts: 12
 - New Subnet: 192.168.1.96/28

- IT:
 - Required Hosts: 10
 - New Subnet: 192.168.1.112/28

Benefits

- Better IP utilization
- Less address wastage
- Supports departmental growth

4. Two devices are configured with the following IPv6 addresses:

- **Device A: 2001:db8::ff00:42:8329/64**
- **Device B: 2001:db8:0:0:1::1/64**

Although both devices are connected to the same network, communication fails.

Task :Debug the IPv6 configuration by:

- Expanding both IPv6 addresses into their complete notation.
- Determining the network prefix of each device.
- Verifying whether both devices belong to the same subnet.
- Identifying the addressing or prefix mismatch.
- Optimizing the IPv6 configuration by assigning appropriate addresses.
- Calculating the number of available host addresses within the corrected subnet.

ANSWER :-

IPv6 Debugging (Same as Q2)

- Device A Expanded: 2001:0db8:0000:0000:0000:ff00:0042:8329
- Device B Expanded: 2001:0db8:0000:0000:0001:0000:0000:0001
- Network Prefix: 2001:db8:0:0::/64
- Same Subnet: Yes
- Optimized Addresses: ::10 and ::20
- Hosts Available: 2^{64}

5. A router contains the following routing table.

Destination Network Next Hop

192.168.1.0/24 10.0.0.2 ;192.168.2.0/24 10.0.0.3 ;Default Route 10.0.0.1

Users report that packets destined for 192.168.2.100 are not reaching the intended network.

Task :Debug the routing configuration by:

- **Identifying the matching routing entry using the longest-prefix match.**
- **Determining the next-hop router selected for packet forwarding.**
- **Verifying whether the destination IP belongs to the intended subnet.**
- **Identifying any routing configuration errors.**
- **Optimizing the routing table to improve packet forwarding efficiency.**
- **Identifying the route that will be used if no matching destination exists.**

ANSWER :-

Routing Table Debugging

Routing Table

- 192.168.1.0/24: Next Hop 10.0.0.2
- 192.168.2.0/24: Next Hop 10.0.0.3
- Default: Next Hop 10.0.0.1

Destination IP: 192.168.2.100

Longest Prefix Match 192.168.2.100 matches 192.168.2.0/24.

Next-Hop Router 10.0.0.3

Destination Verification

- Network: 192.168.2.0/24
- Host: 192.168.2.100
- Status: Valid

Possible Routing Error If packets fail:

- Wrong next-hop IP
- Missing route on downstream router
- Interface to 10.0.0.3 is down

Optimized Routing Table

- 192.168.1.0/24: Next Hop 10.0.0.2
- 192.168.2.0/24: Next Hop 10.0.0.3
- 192.168.3.0/24: Next Hop 10.0.0.4 (future)
- Default: Next Hop 10.0.0.1

If No Match Exists The router uses the Default Route → 10.0.0.1.